

RF-Based Multiple Device Control Using Microcontroller

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Here we describe how to control electrical and electronic gadgets from a remote location

using radio frequency (RF) transmission. An RF interface is used instead of infrared (IR) to avoid the drawbacks of an IR interface. Besides, RF has a longer range. The signal is transmitted



by an RF transmitter and received by an RF receiver to switch on or switch off the desired device. This system can be used to control up to fifteen devices.

Fig. 1 shows the block diagram for RF-based multiple device control using microcontroller. Signals from the keypad are fed to microcontroller AT89C2051, which, in turn, is interfaced to the RF transmitter through encoder HT12E. The microcontroller continuously reads the status of the keys on the keypad.

When any key is pressed, data is passed to the encoder and then to the RF transmitter from where it is transmitted. The RF receiver receives this data and gives it to the RF decoder. The decoder serially converts the serial bit data into four-bit data at a port of microcontroller AT89C51. The microcontroller energises the corresponding relay through a relay driver. Devices are connected to normally-open (N/O) contacts of the relays.

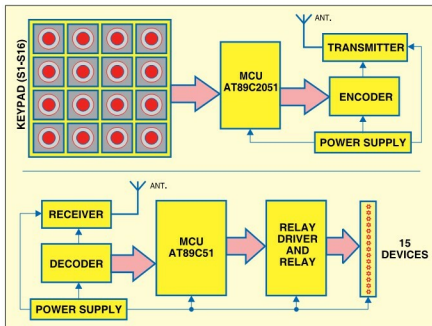


Fig. 1: Block diagram for RF-based multiple device control using microcontroller

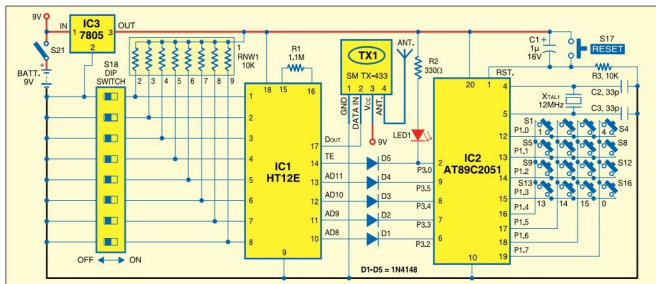


Fig. 2: Transmitter circuit with SM TX-433 RF module (TX1)